Nassau Candy Distributor

Supply Chain Optimization Using Machine Learning

A Data-Driven Factory Reassignment Framework

Internship Research Paper

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Key Results at a Glance	
Dataset	9,783 orders | 14 products | 5 factories
Best Model R^2	0.67 (Random Forest — Medium band)
Avg Lead Time Reduction	20.7 days | 1.51% per viable scenario
Avg Profit Impact	$517 per product (low-risk scenarios)
Recommendation Coverage	99.5% of product revenue

Abstract

This paper presents a data-driven supply chain optimization framework developed during an internship engagement with Nassau Candy Distributor. The study addresses a critical operational challenge: the company's existing static, rule-based approach to factory-product assignment was generating suboptimal shipping routes, excessive lead times, and measurable margin erosion. Using a dataset of 9,783 orders spanning 14 candy products across 5 factories and 4 US/Canadian distribution regions, a five-stage machine learning pipeline was designed and implemented to simulate, quantify, and recommend optimal factory reassignment scenarios.

The pipeline encompasses data preparation and feature engineering, predictive modeling of lead times using three regression algorithms, route and product clustering via KMeans, scenario simulation across all possible factory assignments, and weighted multi-criteria optimization for ranked recommendations. A key analytical contribution was the identification and resolution of a data leakage issue arising from lead time scheduling bands, which led to a per-band modeling approach achieving R^2 values of 0.61–0.67. The framework generated 56 simulated scenarios and identified viable reassignment opportunities covering 99.5% of product revenue, with an average predicted lead time reduction of 20.7 days and a profit impact of $517 per product per year. The findings are surfaced through an interactive Streamlit web dashboard enabling real-time scenario exploration by operations stakeholders.

Keywords: supply chain optimization, machine learning, factory assignment, lead time prediction, KMeans clustering, scenario simulation, gradient boosting, random forest.

1. Introduction

1.1 Background

Supply chain efficiency is a fundamental determinant of profitability for consumer goods distributors. For companies operating multiple production facilities, the decision of which factory manufactures which product — and how that product reaches end customers — has direct implications for shipping costs, delivery lead times, and gross margin. Nassau Candy Distributor operates five factories across the United States and serves customers across four geographic regions: Atlantic, Gulf, Interior, and Pacific.

Historically, Nassau Candy has assigned products to factories using static, legacy rules established when the distribution network was first configured. These rules have not been systematically

reviewed against current demand patterns, regional growth, or operational performance data. As a result, the company faces three interrelated inefficiencies: suboptimal shipping distances where products are manufactured far from the regions they primarily serve; high lead times in specific region-product combinations that are structurally mismatched to their factory; and margin erosion driven by logistics costs that could be reduced through better assignment decisions.

1.2 Problem Statement

The core problem is the absence of any systematic capability to evaluate alternative factory-product configurations before committing to them operationally. Without a simulation framework, Nassau Candy cannot quantify the benefit of a proposed reassignment, compare multiple options against each other, or prioritize action based on business impact. This paper describes the design and implementation of such a framework.

1.3 Objectives

The specific objectives of this research are:

- To perform exploratory data analysis (EDA) on Nassau Candy's order dataset and identify current operational inefficiencies.
- To build predictive models that estimate shipping lead time based on product, factory, region, ship mode, and distance features.
- To cluster routes and products by performance similarity and identify consistently underperforming segments.
- To simulate factory reassignment scenarios for each product and quantify the operational impact across four KPIs.
- To apply a weighted optimization model to rank recommendations and produce an actionable decision support output.
- To deliver an interactive Streamlit dashboard enabling real-time exploration by operations stakeholders.

1.4 Scope

The analysis covers all 14 products in Nassau Candy's portfolio across 5 factories and 4 regions. The dataset spans orders with ship dates projected across 2027–2029. The study focuses on lead time, shipping distance, gross profit, and operational risk as primary decision criteria. Detailed factory capacity constraints and SKU-level demand forecasting are beyond the scope of this analysis, and phased implementation is recommended to manage workload redistribution.

2. Methodology

2.1 Overview of the Pipeline

The analytical framework was implemented as a five-stage Python pipeline. Each stage reads from the output of the previous one, ensuring full reproducibility and clear data lineage. Figure 1 summarizes the pipeline architecture.

Stage	Name	Description	Key Output
1	Data Preparation	Feature engineering, encoding, scaling	nassau_enriched.csv
2	Predictive Modeling	Per-band regression (LR, RF, GBR)	stage2_band_models.pkl
3	Clustering	KMeans routes & products (k=5)	stage3_*_clusters.csv
4	Simulation Engine	56 scenarios, 4 KPI metrics	stage4_simulations.csv
5	Optimization	Weighted scoring, ranked recommendations	stage5_recommendations.csv

Table 1: Five-stage pipeline architecture.

2.2 Stage 1 — Data Preparation & Feature Engineering

The raw dataset comprised 10,194 order records across 18 columns including order date, ship date, product, customer state/province, ship mode, region, sales, units, gross profit, and cost. The following feature engineering steps were applied:

- Lead time computation: calculated as the difference in calendar days between Ship Date and Order Date. The resulting values cluster into three distinct bands corresponding to ship years 2027, 2028, and 2029.
- Factory assignment: each of the 14 products was mapped to its designated factory using the Nassau Candy factory-product correlation table. Five factories were included: Lot's O' Nuts, Wicked Choccy's, Sugar Shack, Secret Factory, and The Other Factory.
- Geographic feature engineering: factory coordinates (latitude, longitude) were assigned from the provided factory location table. Destination coordinates were mapped from US state and Canadian province centroids. Shipping distance was then computed using the Haversine formula, yielding great-circle distances in miles.
- Outlier removal: IQR-based filtering was applied to the four financial columns (Sales, Units, Gross Profit, Cost), removing 411 rows (4.0% of the dataset).
- Encoding: Label Encoding was applied to six categorical columns. StandardScaler normalization was applied to nine numerical columns.

The final enriched dataset contains 9,783 rows and 31 columns (15 raw + 6 encoded + 9 scaled + 1 target), stored as nassau_enriched.csv and used as the master input for all downstream stages.

2.3 Stage 2 — Predictive Modeling

2.3.1 Model Selection

Three regression models were evaluated for lead time prediction: Linear Regression (baseline), Random Forest Regressor (n_estimators=100, max_depth=10), and Gradient Boosting Regressor (n_estimators=100, learning_rate=0.1, max_depth=5). An 80/20 train-test split was used alongside 5-fold cross-validation.

2.3.2 The Scheduling Band Discovery

Initial model training across the full dataset yielded R^2 values approaching zero for all three models. Diagnostic analysis revealed that lead times distribute into three discrete clusters — approximately 908 days, 1,273 days, and 1,638 days — spaced exactly 365 days apart. These correspond to order batches scheduled for delivery in 2027, 2028, and 2029 respectively. The dominant variance in lead time was therefore attributable to the scheduling tier, not to the product, factory, or regional features available in the dataset.

A naive solution of encoding the scheduling band as a feature introduced target leakage (R^2=1.0), since the band was derived directly from the target variable. The correct approach — adopted in this study — was to train one independent model per band, isolating within-band variation where factory, distance, region, and ship mode genuinely matter. This yielded meaningful R^2 values of 0.61–0.67.

2.3.3 Model Performance

Band	Best Model	RMSE (days)	MAE (days)	R^2	CV R^2
Short (~908d)	Gradient Boosting	0.99	0.79	0.636	0.607
Medium (~1273d)	Random Forest	0.98	0.80	0.675	0.668
Long (~1638d)	Gradient Boosting	1.05	0.86	0.609	0.650

Table 2: Model performance by scheduling band. RMSE of approximately 1 day on lead times with a within-band standard deviation of 1.7 days represents strong predictive accuracy.

2.4 Stage 3 — Route & Product Clustering

KMeans clustering (k=5) was applied separately to routes (Factory × Region pairs) and to individual products. The optimal k was selected by maximizing the silhouette score across k=2 to k=6. Route clustering used four features: average lead time, average shipping distance, average gross profit,

and lead time standard deviation. Product clustering used the same feature set aggregated at the product level.

Route clustering achieved a silhouette score of 0.370, and product clustering achieved 0.411. Cluster labels were assigned by ranking cluster centroids on average lead time from Fastest to Slowest. Key findings:

- Sugar Shack → Pacific was identified as the single worst-performing route with an average lead time of 1,517 days — 196 days above the overall mean.
- All four Secret Factory routes clustered into the Slow category, suggesting structural assignment inefficiency.
- Hair Toffee and Lickable Wallpaper were assigned to the Slowest product cluster while simultaneously being the two highest-margin products ($14.88 and $10.00 average gross profit respectively), making them the highest-priority reassignment candidates.

2.5 Stage 4 — Scenario Simulation Engine

For each of the 14 products, reassignment was simulated to each of the 4 alternate factories, generating 56 scenarios in total. For each scenario, the following quantities were computed:

- Simulated shipping distances: recalculated using the Haversine formula from the alternate factory coordinates to each order's destination state centroid.
- Predicted lead times: generated using the appropriate per-band Stage 2 model, with factory encoding and numerical scaling updated for the alternate assignment. A blended estimate was applied: 60% weight to the observed route cluster benchmark and 40% to the model prediction, providing a more stable estimate than either source alone.
- Lead Time Reduction (%): percentage improvement in average lead time versus the current assignment.
- Profit Impact ($): estimated annual cost saving derived from shipping distance reduction, using a distance-based cost proxy of $0.002 per mile per unit.
- Scenario Confidence Score: a composite score (0–1) combining model R^2 (45% weight), route cluster improvement score (30% weight), and order volume adequacy (25% weight).
- Risk classification: scenarios were classified as Low, Medium, or High risk based on order count adequacy, direction of lead time change, and confidence score threshold.

2.6 Stage 5 — Optimization & Recommendation Ranking

All viable scenarios (positive lead time reduction) were ranked using a weighted composite scoring formula:

$$Score = 0.40 \times LT_reduction_norm + 0.35 \times Profit_impact_norm + 0.25 \times Confidence_score - Risk_penalty$$

Each metric was min-max normalized to a [0, 1] scale before weighting. Risk penalties were applied as: Low = 0.00, Medium = −0.05, High = −0.20. The scoring weights were selected to reflect Nassau Candy's operational priorities: lead time reduction is the primary driver (40%), profitability is the secondary driver (35%), and confidence ensures the recommended actions are data-supported (25%). The weights are configurable in the Streamlit dashboard via a real-time priority slider.

3. Exploratory Data Analysis — Key Findings

3.1 Dataset Overview

Total orders (after outlier removal)	9,783
Products	14 (across Chocolate, Sugar, and Other divisions)
Factories	5 (US-based, spanning Arizona to South Carolina)
Distribution regions	4 (Atlantic, Gulf, Interior, Pacific)
Ship modes	4 (Standard Class, Second Class, First Class, Same Day)
Date range	Orders: 2025 | Ship dates: 2027–2029
Average lead time	1,321 days (mean) | Range: 904–1,642 days
Average shipping distance	1,235 miles
Average gross profit	$5.73 per order

3.2 Lead Time Analysis

Lead time varies most significantly by product, with a 205-day spread between the fastest product (Kazookles, 1,251 days) and the slowest (Lickable Wallpaper, 1,456 days). By region, Gulf is the fastest destination (1,314 days average) and Interior is the slowest (1,324 days). Counterintuitively, Standard Class shipping is on average the fastest mode (1,315 days), while First Class is the slowest (1,336 days). This inversion indicates that factory assignment and geographic routing are more influential drivers of lead time than the nominal shipping tier.

3.3 Factory & Distance Analysis

The factory heatmap analysis revealed substantial variation in shipping performance by Factory × Region combination. Lot's O' Nuts, located in Arizona, ships an average of 2,013 miles to Atlantic region customers — the longest route in the portfolio. Wicked Choccy's ships 2,119 miles to Pacific region customers, representing the single longest average distance in the dataset. In contrast, The

Other Factory (Tennessee) achieves distances below 500 miles to the Gulf and Interior regions, making it a natural alternate destination for products currently assigned to more distant factories.

3.4 Profitability Analysis

Gross profit varies substantially by product. Hair Toffee is the highest-margin product at $14.88 average gross profit, followed by Lickable Wallpaper ($10.00), Nutty Crunch ($9.05), and Scrumdiddlyumptious ($9.02). At the lower end, Kazookles ($0.57) and Fun Dip ($1.60) generate minimal margin per order. The combination of high margin and high lead time in Hair Toffee and Lickable Wallpaper — both classified in the Slowest performance cluster — represents the most compelling optimization opportunity in the portfolio.

4. Results

4.1 Clustering Results

Route clustering identified 7 of 20 Factory × Region routes as Slow or Slowest. The Slowest cluster contains a single route — Sugar Shack → Pacific — with an average lead time of 1,517 days, 196 days above the portfolio mean. All four Secret Factory routes fell into the Slow cluster, suggesting that this factory is consistently underperforming relative to its peers regardless of destination region.

Product clustering assigned Hair Toffee and Lickable Wallpaper to the Slowest cluster. The Moderate cluster contains the five Chocolate division products (Fudge Mallows, Milk Chocolate, Nutty Crunch, Scrumdiddlyumptious, Triple Dazzle), all achieving similar lead times in the 1,315–1,329 day range. Kazookles and The Other Factory's products cluster as Fastest, though their low profit margins limit the business value of further optimizing these assignments.

4.2 Simulation Results

The simulation engine generated 56 scenarios across 14 products and 4 alternate factories. Of these, 32 scenarios (57%) produced a positive lead time reduction. High-risk classifications were applied to 34 scenarios, primarily because several Sugar and Other division products have fewer than 10 orders in the dataset, making model predictions unreliable for those specific assignments.

Product	Current Factory	Recommended Factory	LT Reduction (%)	Profit Impact ($)	Score	Risk
Scrumdiddlyumptious	Lot's O' Nuts	The Other Factory	0.35%	$1,398	0.534	Low

Product	Current Factory	Recommended Factory	LT Reduction (%)	Profit Impact ($)	Score	Risk
Nutty Crunch	Lot's O' Nuts	The Other Factory	0.67%	$1,278	0.531	Low
SweeTARTS	Sugar Shack	The Other Factory	5.57%	$5	0.531	Low
Laffy Taffy	Sugar Shack	The Other Factory	5.56%	$5	0.531	Low
Fudge Mallows	Lot's O' Nuts	The Other Factory	0.07%	$1,314	0.508	Low

Table 3: Top 5 ranked recommendations by composite score.

4.3 KPI Summary

KPI 1 — Lead Time Reduction (Operational Gain)	20.7 days avg | 1.51% avg reduction (viable, low-risk)
KPI 2 — Profit Impact (Financial Stability)	$517 avg per product per year (low-risk scenarios)
KPI 3 — Scenario Confidence Score (Reliability)	0.63 avg (0–1 scale, low-risk scenarios)
KPI 4 — Recommendation Coverage (Scalability)	8/14 products | 99.5% of product revenue covered

4.4 Factory Workload Impact

If all top recommendations were actioned simultaneously, The Other Factory would absorb an estimated +9,725 additional orders — a load increase of over 600% from its current baseline. This represents a critical capacity constraint that must be addressed in implementation planning. A phased rollout strategy is strongly recommended, beginning with the highest-scoring Low-risk reassignments and allowing factory operational capacity to adjust before successive phases are activated.

5. Discussion

5.1 Interpretation of Results

The most consistent finding across all analytical stages is the operational advantage of The Other Factory (Memphis, Tennessee) as an alternate production site. Its central geographic position relative to the Gulf and Interior regions — the two largest by order volume — means that reassigning products from coastal factories (Lot's O' Nuts in Arizona, Wicked Choccy's in Georgia) to The Other Factory reduces shipping distances by 200–400 miles on average for affected destinations. This translates into both lead time improvements and logistics cost reductions.

The lead time reduction percentages appear modest at 1–8% across most scenarios. This is partly a consequence of the dataset's scheduling structure: since lead times span two to three years, even a meaningful operational improvement of 20–70 days represents a small percentage of the total. The absolute day reduction is the more actionable metric for operations teams — and a 20-day reduction in lead time for a high-volume product has material implications for inventory planning and customer service levels.

5.2 Limitations

- Factory capacity constraints were not incorporated into the model. The workload analysis flags a significant imbalance risk if all recommendations are actioned together.
- The dataset does not include actual factory scheduling or production lead time data. The lead time recorded represents the full order-to-ship cycle, of which factory production is one component.
- Several products (Hair Toffee, Kazookles, Everlasting Gobstopper) have very low order counts in the dataset, making simulation predictions for those products less reliable. These are flagged as High Risk in the output.
- The shipping cost proxy ($0.002 per mile per unit) is an estimate. Actual logistics cost structures may differ significantly depending on carrier contracts, mode of transport, and volume commitments.
- The Haversine formula computes great-circle (straight-line) distances. Actual road or freight distances may be 15–30% higher depending on routing.

5.3 Future Improvements

- Incorporate factory capacity data: adding production capacity constraints would allow the optimization model to respect operational limits and produce implementable plans without manual phasing.
- Time-series demand forecasting: integrating seasonal demand patterns would allow the framework to recommend dynamic factory assignments that shift across quarters rather than static reassignments.

- Actual distance data: replacing Haversine estimates with Google Maps Distance Matrix API data would improve the accuracy of shipping distance calculations and cost estimates.
- Multi-objective optimization: replacing the weighted scoring approach with Pareto-frontier optimization (e.g., using scipy.optimize) would surface the trade-off curve between speed and profit more precisely.
- Expanded product catalog: the current analysis covers 14 products and 5 factories. The pipeline is fully modular and can scale to larger product catalogs without structural changes.

6. Conclusion

This research demonstrates that a systematic, data-driven approach to factory-product assignment can identify meaningful operational improvement opportunities that legacy static rules overlook. By combining predictive modeling, unsupervised clustering, scenario simulation, and weighted optimization into a reproducible five-stage pipeline, the study delivers both analytical depth and practical decision support.

The framework identified 8 out of 14 products with viable, low-risk reassignment opportunities — covering 99.5% of product revenue — and ranked them by a composite score that balances lead time reduction, profit impact, and prediction confidence. The most impactful recommendations center on migrating production from the geographically isolated Lot's O' Nuts (Arizona) and Sugar Shack (Minnesota) facilities to The Other Factory (Tennessee) for products that primarily serve Gulf and Interior region customers.

The accompanying Streamlit dashboard makes these findings accessible to operations stakeholders without requiring any code interaction. The real-time priority slider enables decision-makers to explore how recommendations shift when the business emphasis moves between speed and profitability — ensuring the framework adapts to Nassau Candy's strategic priorities rather than imposing a single fixed view.

The primary recommendation for implementation is a phased approach beginning with the two or three highest-scoring reassignments, monitoring operational performance for 60–90 days, and then proceeding to subsequent phases. This approach avoids the capacity shock that simultaneous execution of all recommendations would impose on The Other Factory, while still capturing the majority of the efficiency gains identified in this study.

7. References & Tools

The following tools, libraries, and methodologies were used in this study:

- Python 3.11 — primary programming language
- pandas, numpy — data manipulation and numerical computing
- scikit-learn — machine learning models (LinearRegression, RandomForestRegressor, GradientBoostingRegressor, KMeans, StandardScaler, LabelEncoder)
- matplotlib, seaborn — exploratory data analysis visualization
- Streamlit — interactive web dashboard framework
- Plotly — interactive chart rendering within Streamlit
- Haversine formula — great-circle distance computation (implemented natively, no external library)

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Appendix A — Factory Reference Data

Factory	Location	Latitude	Longitude
Lot's O' Nuts	Arizona, USA	32.8819	-111.7680
Wicked Choccy's	Georgia, USA	32.0762	-81.0884
Sugar Shack	Minnesota, USA	48.1191	-96.1812
Secret Factory	Illinois, USA	41.4463	-90.5655
The Other Factory	Tennessee, USA	35.1175	-89.9711

Appendix B — Product-Factory Assignment

Division	Product Name	Current Factory
Chocolate	Wonka Bar - Nutty Crunch Surprise	Lot's O' Nuts
Chocolate	Wonka Bar - Fudge Mallows	Lot's O' Nuts
Chocolate	Wonka Bar - Scrumdiddlyumptious	Lot's O' Nuts
Chocolate	Wonka Bar - Milk Chocolate	Wicked Choccy's
Chocolate	Wonka Bar - Triple Dazzle Caramel	Wicked Choccy's
Sugar	Laffy Taffy	Sugar Shack
Sugar	SweeTARTS	Sugar Shack
Sugar	Nerds	Sugar Shack
Sugar	Fun Dip	Sugar Shack
Other	Fizzy Lifting Drinks	Sugar Shack
Sugar	Hair Toffee	The Other Factory
Other	Lickable Wallpaper	Secret Factory
Other	Wonka Gum	Secret Factory
Other	Kazookles	The Other Factory